

Database Schema Design

Topology: Shared Database, Shared Schema.

Multi-tenancy: Enforced via `company_id` on all resource tables.

Naming Convention: `snake_case` (all lowercase, underscores for spaces).

1. users

Column Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for the user.
email	VARCHAR(255)	Unique, Not Null	User's email address (login).
hashed_password	VARCHAR(255)	Not Null	Bcrypt hashed password.
created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp.

2. companies

Column Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for the tenant.
name	VARCHAR(255)	Not Null	Name of the tenant/company.
created_at	TIMESTAMP	DEFAULT NOW()	Company creation timestamp.

3. company_members

Column Name	Data Type	Constraints	Description
user_id	UUID	FK -> users.id	Links user to company.
company_id	UUID	FK -> companies.id	Links company to user.
role	ENUM	Not Null	'owner', 'admin', 'member', 'viewer'.
Note		UNIQUE(u_id, c_id)	Prevents duplicate memberships.

4. folders

Column Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for the

			folder.
company_id	UUID	FK -> companies.id	Tenant Isolation Key.
parent_id	UUID	FK -> folders.id	Allows nested folder structures.
name	VARCHAR(255)	Not Null	Folder name.
deleted_at	TIMESTAMP	Nullable	For soft-delete functionality.

5. documents

Column Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for the document.
company_id	UUID	FK -> companies.id	Tenant Isolation Key.
folder_id	UUID	FK -> folders.id	Folder location (nullable for root).
filename	VARCHAR(255)	Not Null	Sanitized name of file.
file_type	VARCHAR(50)	Not Null	PDF, DOCX, TXT, MD, etc.
status	ENUM	DEFAULT 'pending'	pending, processing, completed, failed.
uploaded_by	UUID	FK -> users.id	User who uploaded the file.
created_at	TIMESTAMP	DEFAULT NOW()	Upload timestamp.
deleted_at	TIMESTAMP	Nullable	For soft-delete functionality.

6. document_chunks (AI Search)

Column Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier for the chunk.
company_id	UUID	FK -> companies.id	Tenant Isolation Key.
document_id	UUID	FK -> documents.id	Link to parent document.
chunk_text	TEXT	Not Null	Extracted text snippet.
chunk_index	INTEGER	Not Null	Order of the chunk in the document.
embedding	VECTOR(768)	Not Null	Generated AI vector.

7. audit_logs

Column Name	Data Type	Constraints	Description
id	UUID	PK	Unique identifier.
company_id	UUID	FK -> companies.id	Tenant Isolation Key.
user_id	UUID	FK -> users.id	Actor performing the action.
action	VARCHAR(100)	Not Null	e.g., 'DOC_UPLOAD', 'SEARCH'.
ip_address	VARCHAR(45)	Nullable	IP address of requester.
timestamp	TIMESTAMP	DEFAULT NOW()	Action occurrence time.